

Phlebotomine sandflies (Diptera: Psychodidae) of Marrakech city, Morocco

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Received 9 February 2007, Revised 20 March 2007,

Accepted 23 March 2007

The sandflies at 17 sites within or near the Moroccan city of Marrakech were surveyed between 2002 and 2006. Overall 2310 specimens were collected and five species were identified. *Phlebotomus* (*Phlebotomus*) *papatasi* was the most prevalent species (47.4%), followed by *Sergentomyia* (*Sergentomyia*) *minuta* (16.6%), *S. (S.) fallax* (16.1%), *P. (Paraphlebotomus) sergenti* (12.8%) and *P. (Larroussius) longicuspis* (7.1%). *Phlebotomus sergenti* appeared to be the most endophilic species. When the collection sites were categorized into five types of habitat, only *P. papatasi* and *S. minuta* were found to be common in every habitat type. Urbanization seemed to have a significant effect on the numbers and species of sandfly. In the year after the demolition and redevelopment of the Akioud neighbourhood, there were far fewer sandflies and species of sandfly than observed before the redevelopment. Two years later, however, the sandfly population was found to be much larger than that seen before the neighbourhood was demolished.

Phlebotomine sandflies (Diptera: Psychodidae) are obligate vectors of the parasitic protozoa, of the genus *Leishmania* (Kinetoplastida: Trypanosomatidae), that cause leishmaniasis. Although there is a long history of human leishmaniasis in Morocco (Colonieu, 1931), the disease's impact on human health in this part of Africa was not really recognised until the 1970s. Since then, Moroccan leishmaniasis has been the subject of an eco-epidemiological programme directed by Professor J. A. Rioux (Rioux, 2001).

In 2003, according to the national Ministry of Health, 3152 cases of human cutaneous leishmaniasis (CL) caused by

L. major (2140 cases) or *L. tropica* (1012) and 115 cases of human visceral leishmaniasis (VL) caused by *L. infantum* were reported in Morocco (Anon., 2003). Recently, the local distribution of these forms of the disease appears to have changed, with increasing numbers of cases reported from the peri-urban and urban areas where the disease was once unknown or relatively rare. In 2004, for example, 23 cases of CL and seven of VL were reported in the prefecture of Marrakech (Anon. 2004). For those involved in the epidemiology of leishmaniasis, Marrakech is an interesting study site because it lies close to the focus of zoonotic cutaneous leishmaniasis (ZCL) in Ouarzazate (Rioux *et al.*, 1986) and the foci of anthroponotic cutaneous leishmaniasis (ACL) in Azilal (Pratlong *et al.*, 1991) and Chichaoua (Guernaoui *et al.*, 2005). Rioux *et al.* (1986, 1997) classified the area of Marrakech as being at

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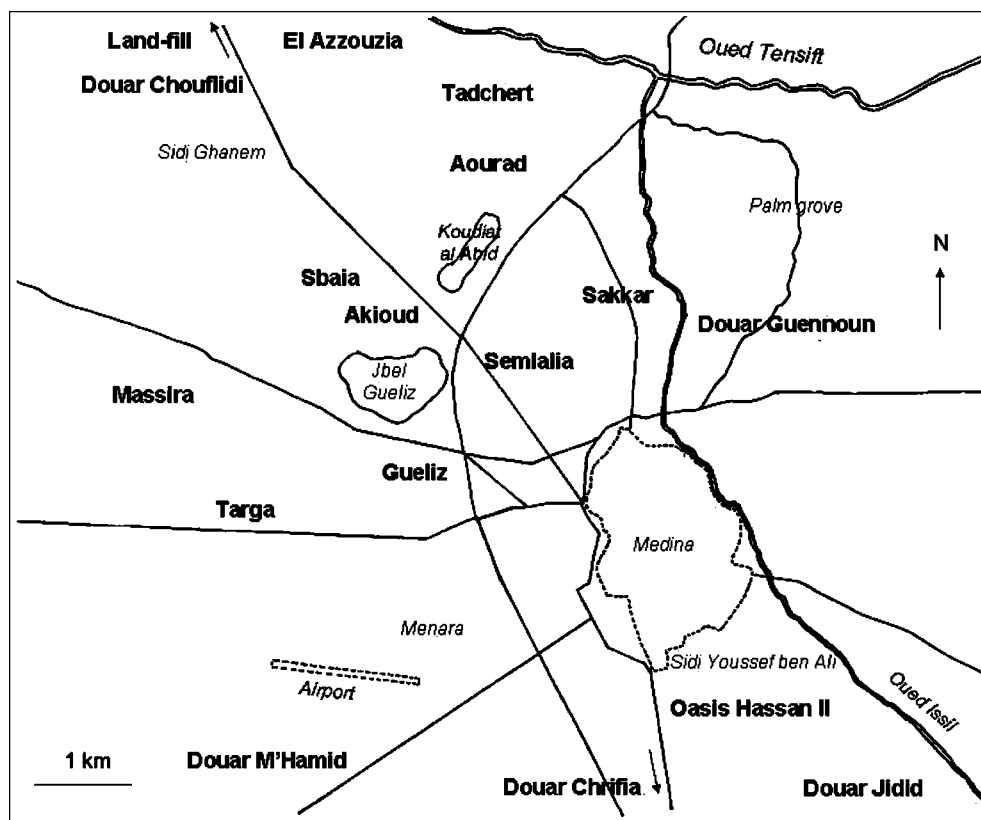


FIG. 1. Sketch map of Marrakech city, showing the location of the 17 study sites.

risk of *L. major* transmission (because of its arid climate). In addition, *Phlebotomus papatasi* Scopoli, the proven vector of *L. major* in Morocco (Rioux *et al.*, 1986), is now known to be active throughout the year in the urban area of Marrakech (Boussaa *et al.*, 2005).

Marrakech, like most of Morocco's large cities, has expanded quickly over recent years. Mass emigration to the city from the south of Morocco has not only fueled this expansion but led to the urbanization of peripheral, rural areas and the development of many unplanned extensions to villages (*douars*) with poor facilities and sanitation. The spread of human leishmaniasis into urban areas is probably attributable to this partial urbanization of once rural areas of the prefecture (WHO, 2002; Desjeux, 2004).

The aim of the present study was to explore the interaction between the

environmental changes taking place in the peri-urban areas of Marrakech and the sandfly fauna, as an initial step in the development of effective strategies for the control of leishmaniasis in the prefecture.

MATERIALS AND METHODS

Study Sites

The 17 collection sites (Fig. 1) all lie within or close to the city of Marrakech (31°36'N, 8°02'W), which lies, at an altitude of about 470 m above sea level, on the El Haouz plain, just to the north of the foothills of the High Atlas chain. The plain is only broken by the chalky hills of Jbel Gueliz and Koudiat Al Abid. The climate is arid. The mean monthly temperatures range from 41.1°C (in July) to just 4°C (in December).

Habitat Categories

In order to quantify the effect of urbanization on sandfly distribution, the 17 study sites were categorized into five types of habitat: 'intensely urban' areas; areas next to *douars*; *douars* in urban areas; *douars* in suburban areas; and waste-management sites.

Four sites, all within the modern city, were considered to be 'intensely urban'. These were the central business district (Gueliz) and three areas of housing development that was very well organized, with private houses with gardens (Targa), public-sector housing (flats) without gardens but with tree-lined streets (Massira), or a mix of private and public-sector housing with some gardens (Semlalia).

The two study areas close to the *douars* — Sakkar (close to Douar Guennoun) and Sbaia (close to Douar Akioud) — have well kept private houses surrounded by gardens.

Although the *douars* that were investigated lie in urban (Guennoun, Jidid, Akioud and Oasis Hassan II) or suburban areas (Aourad, M'hamid, Tadchirt, Chrifia and Choufliidi), the *douars* themselves are predominantly rural in character, with stables adjacent to the human dwellings and poor sanitary conditions.

The two waste-management sites investigated were the wastewater-spreading area of Marrakech (El Azzouzia) and a land-fill site that lies 15 km to the north-north-west of the city.

Entomology

Entomological surveys were carried out at most of the sites in June 2002, June 2004, August 2005 and August 2006 (see Table). Additional collections were made at Semlalia and Guennoun in June 2003 and at Akioud in May–August 2003 and May–August 2004. At each site at each time-point, about 40 sticky traps (each a white or yellow, A5 sheet of paper coated with castor oil) were set overnight (18.00–09.00 hours). At the sites with houses, such traps were

placed both inside and outside the human dwellings.

The sandflies caught on the traps were preserved in 70% ethanol, cleared in 20% KOH and Marc-André solution (chloral hydrate/acetic acid), then dissected and mounted in Hoyer's medium. They were then identified to species level by morphological examination of the male genitalia and the female spermathecae and pharynges.

RESULTS

Overall, 2310 sandflies of five species were collected (see Table). *Phlebotomus* was the predominant genus, representing 67.3% of the specimens caught. *Phlebotomus* (*Phlebotomus*) *papatasi* was the predominant species (47.4%), followed by *Sergentomyia* (*Sergentomyia*) *minuta* Rondani (16.6%), *S. (S.) fallax* Parrot (16.1%), *Ph. (Paraphlebotomus) sergenti* Parrot (12.8%) and *Ph. (Larroussius) longicuspis* Nitzulescu (7.1%). As usually observed in collections made with sticky traps, more males than females of each species were caught (Table). The numbers of sandflies caught on the yellow traps were similar to those caught on the white traps ($P=0.6$; data not shown).

The type of habitat appeared to have a significant effect on sandfly abundance ($\chi^2=256$; $P<0.0001$) — with >70% of all the sandflies collected coming from the *douars* — and on the species of sandfly present. In the 'intensely urban' sites, for example, no *S. fallax* were caught and *Ph. sergenti* was relatively rare (see Table and Figure 2).

When the numbers of sandflies caught inside human dwellings were compared with those caught outside (Fig. 3), both species of *Sergentomyia* appeared to be exophilic. The *Phlebotomus* species in general and the female *Phlebotomus* in particular tended to be more endophilic. Although 60% of the *Phlebotomus* females were collected inside, 78% of the *Sergentomyia* females were collected outside.

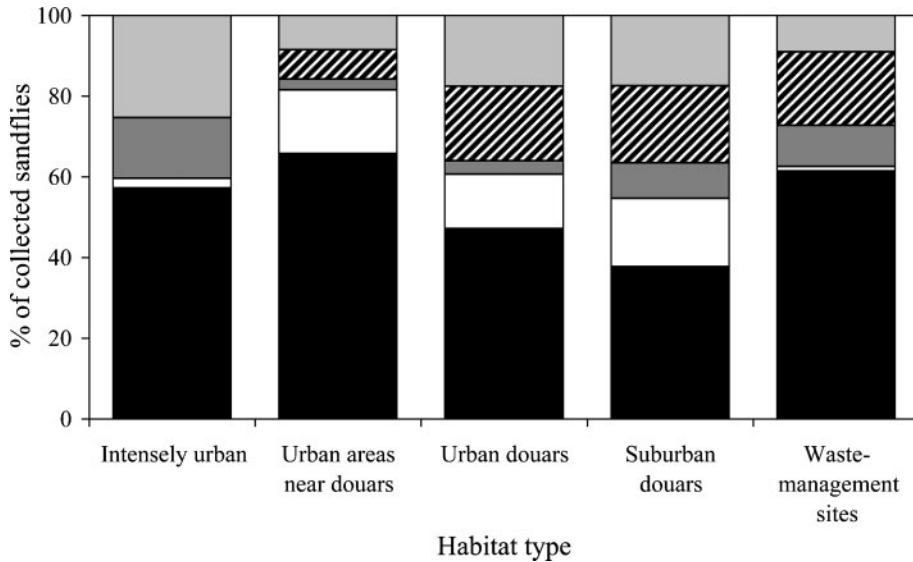


FIG. 2. The relative abundances of *Phlebotomus papatasi* (■), *Ph. sergenti* (□), *Ph. longicuspis* (■), *Sergentomyia fallax* (▨) and *S. minuta* (■) in each category of habitat sampled in Marrakech.

In the June of each year from 2002 to 2005, more sandflies were caught in the *douar* of Guennoun than in the intensely urban Semlalia (Fig. 4). There was no significant correlation between the minimum ($r=0.2$), mean ($r=0.06$) or maximum ($r=-0.05$) temperatures recorded and the total numbers of sandflies collected at these two sites (Fig. 4).

During the study period, the ongoing programme of redevelopment in Marrakech led to the demolition and clearance of the *douar* of Akioud in early 2004. A comparison of the results of the entomological surveys in May–August 2003 (i.e. before the demolition) with those of the surveys conducted in May–August 2004 (i.e. after the demolition) indicates that the

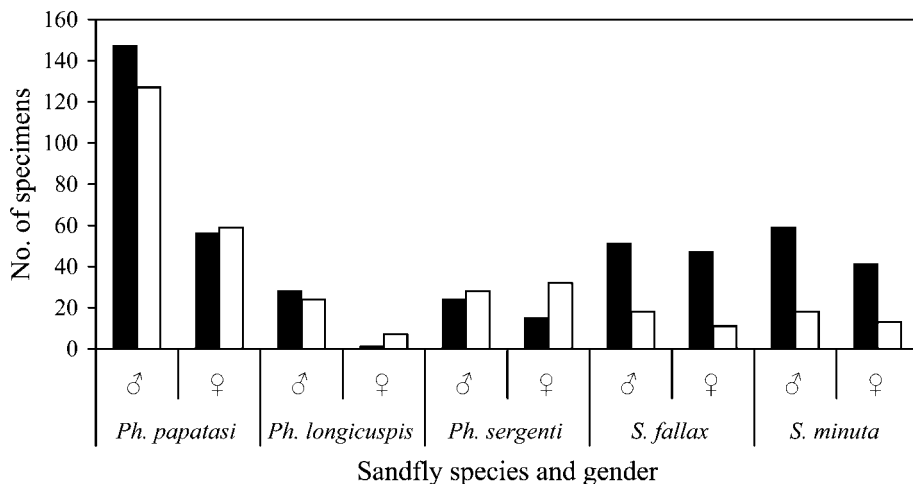


FIG. 3. The numbers of male and female sandflies collected outside (■) and inside (□) human dwellings in the urban areas of Marrakech city.

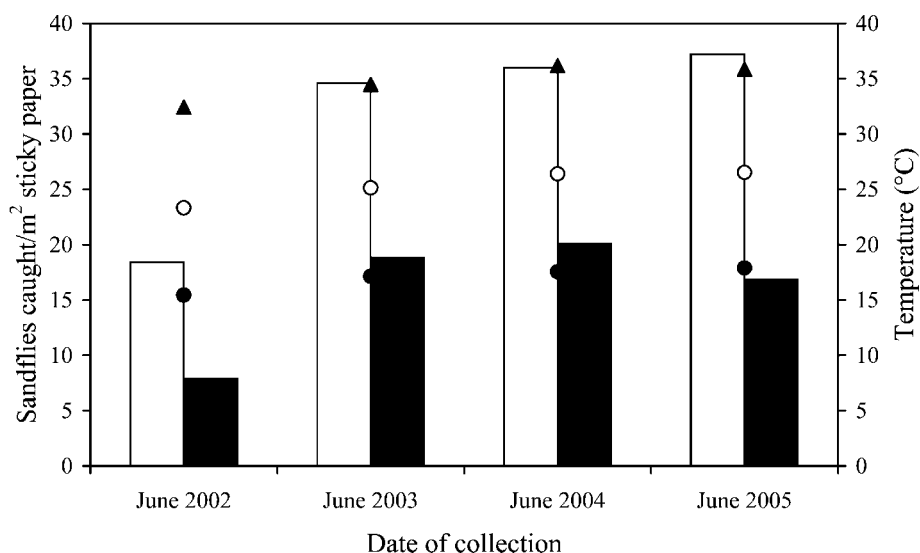


FIG. 4. The mean numbers of sandflies caught/m² of sticky paper in Guennoun (□) and Semlalia (■), and the minimum (●), mean (○) and maximum (▲) temperatures recorded in Marrakech, in the month of June in 2002, 2003, 2004 and 2005.

demolition rapidly led to a fall in sandfly abundance, with the apparent disappearance of all female sandflies (Fig. 5). By August 2006, however, sandfly densities in Akioud appeared much higher than those recorded pre-demolition (Fig. 5) and both sexes of all four species recorded pre-demolition (*Ph. papatasi*, *Ph. sergenti*, *S. fallax* and *S. minuta*) were again present.

DISCUSSION

Sandfly Populations

Of the five species of phlebotomine sandfly caught during the present study, two are the proven vectors of parasites that can cause CL, *Ph. papatasi* transmitting *L. major* and *Ph. sergenti* transmitting *L. tropica* (Killick-Kendrick, 1990). As the best choice of a vector-control strategy is dictated by sandfly behaviour, the level of endophily of each species was investigated. Although similar numbers of specimens of *Phlebotomus* were collected inside and outside human dwellings, most of the female *Phlebotomus* collected were caught inside. As Oliveira *et al.*

(2004) also observed, the populations of the vector species inside households appeared to be predominantly female whereas those in peri-domestic areas had similar numbers of both genders. In such conditions, most transmission of leishmanial parasites to humans probably occurs inside houses. In urban and suburban areas of Brazil, Gomes *et al.* (1986) caught 10-fold more sandflies inside of dwellings than in peri-domestic environments.

Phlebotomus sergenti appeared to be the most endophilic of the sandfly species caught in the present study (with 60% of all specimens caught inside). This species also showed a 'preference' for domestic or peridomestic habitats in Chichaoua, Morocco (Guernaoui *et al.*, 2005), and was mostly associated with domestic habitats in the peri-urban and urban areas of Italy investigated by D'Urso *et al.* (2004). ACL is restricted to the Old World, where most of the foci are urban or suburban and caused by *L. tropica* transmitted by *Ph. sergenti* (Desjeux, 2001). *Phlebotomus sergenti* is probably the primary vector in the focus of ACL discovered, in 1995, in urban and

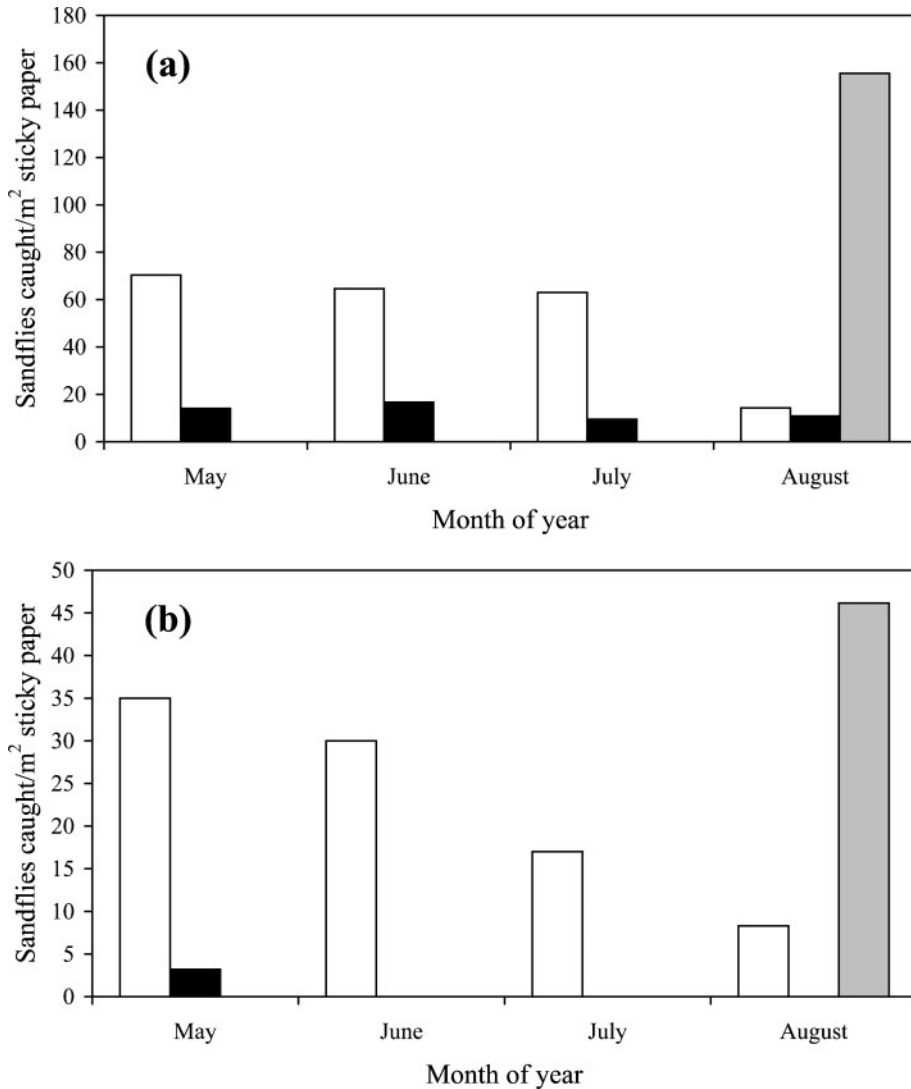


FIG. 5. The mean numbers of sandflies (a) and of female sandflies (b) caught/m² of sticky paper in Akioud, in May–August before (2003; □), just after (2004; ■) and 2 years after (2006; ■) the demolition and clearance of the *douar*.

peri-urban areas of the province of Taza, in north-eastern Morocco (Guessous-Idrissi *et al.*, 1997).

Urbanization Effect

In Marrakech, sandfly abundance appeared to decrease with increasing urbanization, the areas close to *douars* holding larger populations than the 'intensely urban' areas but smaller populations than the *douars* themselves (see Table). The incineration of

rubbish and low numbers of humans at the landfill site may have kept sandfly numbers there relatively low. The *douars* are unplanned communities with poor sanitation and large numbers of emigrants from southern Morocco. The conditions in the *douars* may favour sandfly breeding and the emigrants may be carrying leishmanial infections from more rural areas of the country. The *L. tropica* MON-102 that, in 2002, caused many cases of CL in the

Moroccan city of Zouagha My Yacoub was probably brought to the city by one or more emigrants from rural areas (Rhajaoui *et al.*, 2004).

In the present study, *Ph. papatasi* was the predominant sandfly at every site. It and *S. minuta* appeared to be the species least affected by habitat type (although, in the short-term, female *Ph. papatasi* were made rare in Akioud by the demolition of the dwellings). Papadopoulos and Tselentis (1994) detected *Ph. papatasi* in most of their collection sites in urban areas of Athens, and this species appears to adapt well to anthropogenic environments. *Phlebotomus longicuspis*, a suspected vector of *L. infantum* (Killick-Kendrick, 1990), was only common in one *douar* (Aourad) and, more surprisingly, in 'intensely urban' Semlalia. It may be the gardens in Semlalia that provide enough humidity, detritus, and breeding sites to support the breeding of *Ph. longicuspis*. No *S. fallax* were found at the most urban sites and this species was only common in the *douars*, perhaps because the *douars* have relatively large populations of the reptiles on which this species, and other *Sergentomyia*, feed. Similarly, *Ph. sergenti* was rare in the 'intensely urban' sites and relatively common in the *douars*, where conditions are more rural. In southern Anatolia, Turkey, the highest incidence of ACL is recorded in the poorest suburbs, where cattle are commonly kept in the basements of houses. These damp basements and the cow dung in them probably provide good conditions for sandflies, including the primary vector (*Ph. sergenti*), to breed (WHO, 2002).

It is unclear why the catches in Semlalia and Guennoun were much lower in June 2002 than in the same month of 2003, 2004 or 2005 (Fig. 4). The difference appears unrelated to temperature. There are several possible reasons why the catches in Guennoun were always greater than in Semlalia. Guennoun is a *douar* with poor sanitation and many rodents and dogs, whereas Semlalia is a well organized and

clean urban area with good sanitation and a human community that is rich enough to use insecticides. In Colombia, cases of ZCL have recently been reported from several urban areas, and these have been linked to an urban transmission cycle that is thought to have been triggered by unplanned human settlements on the edges of the urban areas (WHO, 2002). As seen in the deserts of Israel, anthropogenic disturbance can not only lead to larger populations of sandflies but also to changes in the spatial distribution of the 'reservoir' hosts of leishmanial parasites (Wasserberg *et al.*, 2003). Over the last decade, cases of CL have been reported in Ouagadougou, the capital city of Burkina Faso, and linked to public works in the new neighbourhood of 'Ouaga 2000' (Traoré *et al.*, 2001). In North Africa, the increasing incidence of canine leishmaniasis in Algiers has been attributed to the mobility of the human population and unplanned urbanization (Harrat and Belkaid, 2002).

The demolition and subsequent development of the *douar* of Akioud during the present study allowed the effects of such a change on the local sandfly population to be explored. In the short-term, the demolition led to a drop in the size and variety of the sandfly population. Teodoro *et al.* (2003) reported a similarly dramatic drop in the size of the sandfly population in the Brazilian city of Doutor Camargo, following redevelopment, the cleaning of peridomestic areas, and the 'disinsectization' of buildings. In Jordan, Kamhawi *et al.* (1991) showed that, although development and construction activities initially decreased numbers of sandflies, the local sandfly populations were able to recover and often expand beyond their pre-development sizes. Such recovery and expansion were recorded in Akioud, where the sandfly population 2 years after the demolition and clearance appeared several-fold larger than that observed (at a similar time of year) pre-development. The short-term effects of the redevelopment of Akioud appeared to be more severe on the female *Phlebotomus* than on the male, perhaps

because the human population (presumably used as a bloodmeal source by the females) was temporarily depleted and/or because there had been a large emigration of the females (Killick-Kendrick *et al.*, 1984).

Changing environmental conditions caused by habitat destruction, such as those associated with the process of urbanization, often have an enormous influence on vector populations and, consequently, on the epidemiology of vector-borne disease. While some species of potential vector may disappear, others may become more abundant. In one study in Brazil, for example, *Psychodopygus intermedius* was apparently favoured by anthropic alteration while the process of urbanization seemed to be unfavorable to *Lutzomyia migonei* and *Pintomyia fischeri* (Gomes *et al.*, 1986). In Marrakech, however, *Ph. papatasi* appears to be relatively unaffected by urbanization. Worryingly, the improvement of poor communities such as those in the *douars* may lead to significant increases in the numbers of *Ph. papatasi*, the primary vector of *L. major* in Morocco. There is a clearly a need for careful co-ordination between the health, development and other sectors of the Moroccan government, and for continuous surveillance of the vector populations in urban as well as rural settings.

ACKNOWLEDGEMENTS. The authors thank the Regional Service of the Moroccan Ministry of Health (SIAAP Menara-Gueliz, *Laboratoire Régional de Diagnostic Épidémiologique et d'Hygiène du Milieu Marrakech*) for help and epidemiological data. They are also very grateful to S. Mahamdallie (Molecular Systematics Laboratory, Entomology Department, Natural History Museum, London) for her critical review of the manuscript and useful discussions.

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